

FINAL YEAR PROJECT STATUS REPORT

Advanced AI Reasoning System Pro

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Submission date: April 5, 2026

Reporting period: February 1, 2026 to April 4, 2026

Project type: Final Year Project

Deployment platform: Hugging Face Spaces

Executive Summary

The **Advanced AI Reasoning System Pro** is an AI-based application designed to implement and demonstrate multiple structured reasoning strategies using large language models. During the present reporting period, the project progressed from initial setup into a working prototype that includes backend reasoning logic, model integration, conversation handling, export support, and a Gradio-based user interface.

The application is currently **developed, deployed, and demonstrable**, but it is **not yet fully stable**. While the core workflow is functional in development, the deployed version still shows API-related issues. In addition, the project was affected by changes in Groq model availability, including deletion and deprecation of several previously listed models. UI inconsistencies and integration bugs also remain.

Thus, the present status of the project is best described as **working in core functionality, but still under debugging, stabilization, and final refinement**.

1. Project Objective

The project aims to build an interactive AI reasoning platform through which a user can:

- submit prompts using a web interface
- apply different reasoning methodologies to a problem
- compare structured AI-generated outputs
- manage conversation history and exports
- observe practical challenges involved in deployment of LLM-based systems

The broader academic objective of the project is to study how structured reasoning methods can improve AI response quality while also addressing the engineering challenges of making such a system publicly usable.

2. Progress Overview

2.1 Work Completed

Area	Status	Remarks
Project structure and setup	Completed	Modular project structure, environment setup, and configuration prepared
Backend reasoning pipeline	Completed	Core reasoning and conversation flow implemented
Groq API integration	Completed with issues	Integration works, but model availability changes caused instability
Export functionality	Completed	Support added for Markdown, text, JSON, and PDF-related export
Gradio user interface	Completed with issues	Functional UI created, but refinement is still required
Deployment on Hugging Face Spaces	Completed with issues	Deployment done, but deployed app currently shows API errors
Testing and debugging	Ongoing	Stability improvements still in progress

2.2 Visual Progress Snapshot

Backend reasoning engine	[#####]	90%
API integration	[#####]	70%
User interface	[#####]	75%
Export functionality	[#####]	90%
Deployment setup	[#####]	65%
Testing and debugging	[#####]	55%
Overall project completion	[#####]	75%

2.3 Issue Distribution

Deployment / API issues	[####]	40%
UI / UX issues	[###]	25%
Model compatibility issues	[##]	20%

General bugs / edge cases [##] 15%

3. Major Progress Achieved

3.1 Backend and Reasoning System

The backend architecture has been substantially developed. Core modules for reasoning flow, prompt handling, conversation tracking, export services, and related utilities have been implemented. This gives the project a clear and modular base for further refinement.

3.2 AI Reasoning Features

The project includes multiple reasoning-oriented workflows rather than a simple single-response chatbot design. This is a major strength of the project from an academic point of view, since it focuses on structured reasoning behavior and comparative AI output generation.

3.3 User Interface Development

A Gradio-based interface has been developed to provide an accessible front end for user interaction. The system is usable for demonstration purposes, but UI consistency, spacing, alignment, and some feedback mechanisms still need improvement.

3.4 Deployment Achievement

The project has successfully reached the deployment stage on **Hugging Face Spaces**, which marks an important milestone. However, the deployed version is still not fully reliable because API-related errors are appearing during execution.

4. Current Issues and Limitations

4.1 Deployment API Errors

Although the application is deployed, the hosted version currently shows API errors. This affects the reliability of the online version and indicates that deployment-side configuration and runtime handling still need further debugging.

4.2 Groq Model Deletions and Deprecations

At the beginning of the project, Groq offered a broader set of listed models. During the course of development, several of these models were removed, dep-

recated, or changed in availability. This directly affected the project because model configurations that were initially valid became outdated later.

As a result:

- some model references required revision
- deployment behavior became less predictable
- extra debugging effort was needed to maintain compatibility

This was a significant external challenge that impacted project stability.

4.3 UI Issues

The interface is functional, but the following improvements are still needed:

- better alignment and spacing in some sections
- improved consistency of visual presentation
- clearer display of runtime and API errors
- final polishing for submission/demo quality

4.4 Remaining Bugs

Some bugs are still present in the integrated system, especially where backend processing, API communication, and UI handling interact together. These issues do not invalidate the work completed so far, but they do show that final stabilization is still pending.

5. Present Status of the Project

The present project status may be summarized as follows:

- the main concept has been successfully implemented
- the application is working in core functionality
- the project has been deployed publicly
- the deployed version currently produces API errors
- Groq model deprecations affected implementation consistency
- UI refinement and debugging are still in progress

Therefore, the project is in the **working prototype and stabilization stage**. It has clearly advanced beyond the initial phase, but final refinement is still required before it can be considered fully complete and stable.

6. Planned Work

The next phase of work will focus on stabilization and refinement:

1. Resolve API errors in the deployed Hugging Face Spaces version.

2. Update model configuration based on currently supported Groq models.
 3. Improve user interface consistency and feedback handling.
 4. Fix remaining integration and runtime bugs.
 5. Perform final testing for reliable demonstration and submission.
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7. Learning Outcomes

This project has provided practical exposure to:

- integration of LLM APIs in a real application
- implementation of structured AI reasoning methods
- modular software design for AI systems
- deployment issues in hosted environments
- adaptation to changing external API/model ecosystems

These outcomes are academically valuable because they reflect both technical development and real-world software engineering challenges.

8. Conclusion

In conclusion, the **Advanced AI Reasoning System Pro** has achieved substantial progress during the current reporting period. The project architecture, reasoning workflow, interface, and deployment pipeline have all been established. The project is meaningful, demonstrable, and technically relevant as a final year project.

At the same time, some important issues remain unresolved, especially deployment API errors, Groq model deprecations, UI refinement, and remaining bugs. Hence, the most accurate status statement is:

The project is developed and deployed, with core functionality working, but it is still undergoing debugging, stabilization, and final improvement.